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Fluorescence recovery protein: a powerful yet underexplored regulator of photoprotection in cyanobacteria†

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Cyanobacteria utilize an elegant photoprotection mechanism mediated by the photoactive Orange Carotenoid Protein (OCP), which upon binding dissipates excess energy from light-harvesting complexes, phycobilisomes. The OCP activity is efficiently regulated by its partner, the Fluorescence Recovery Protein (FRP). FRP accelerates OCP conversion to the resting state, thus counteracting the OCP-mediated photoprotection. Behind the deceptive simplicity of such regulation is hidden a multistep process involving dramatic conformational rearrangements in OCP and FRP, the details of which became clearer only a decade after the FRP discovery. Yet many questions regarding the functioning of FRP have remained controversial. In this review, we summarize the current knowledge and understanding of the FRP role in cyanobacterial photoprotection as well as its evolutionary history that presumably lies far beyond cyanobacteria.

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Introduction

Non-photochemical quenching (NPQ) mediated by the photoactive and water-soluble Orange Carotenoid Protein (OCP) is at the front line of the fight against the photodamage in cyanobacteria, which prosper also under high light conditions (for earlier reviews, see[1–8]). OCP bears a keto-carotenoid molecule and is an efficient singlet oxygen quencher.[9] Probably more important is OCP's light-controlled ability to decouple energy transfer from specific cyanobacterial light-harvesting complexes – phycobilisomes (PB)[10] – to reaction centers in order to

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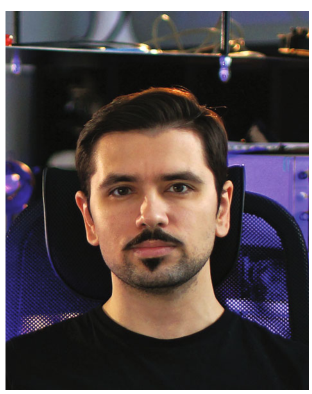

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reduce the efficiency of photosynthesis under potentially harmful conditions. The photoactive properties of OCP, uniquely utilizing ketocarotenoid as a blue light sensor[11] and combining both receptor and effector functions in one protein,[12] justify the significant attention of researchers, which is reflected in more than 100 articles published over the last decade. Not surprisingly, such a peculiar photoreceptor overshadowed the second, colorless participant in the photoprotection process, the so-called Fluorescence Recovery Protein (FRP), which effectively downregulates OCP.[13] Numerous reviews have been primarily dedicated to OCP, but despite the rapidly accumulating knowledge about the functioning of FRP, there is no independent review on FRP to date. We intend to fill this gap by reviewing the available literature on the structure and function of this remarkable protein in the context of the regulation of photoprotection in cyanobacteria. Based on the phylogenetic analysis of FRP-like sequences currently available for different groups of bacteria and archaea, we propose the existence of other yet unknown FRP functions beyond those associated with cyanobacterial photoprotection.

Structure, function and photoactivity of OCP

Since FRP was discovered due to its effect on the OCP-mediated NPQ and OCP photocycle,[13] it is reasonable to start from a brief description of the structure and function of OCP. OCP is composed of two structurally different domains: an all-α N-terminal domain (NTD) and a mixed α/β C-terminal domain (CTD),[14] which are stabilized by the interdomain interface and an N-terminal extension (NTE) covering the CTD (Fig. 1A). A single ketocarotenoid molecule is important for structural stabilization and photoactivity of OCP; it spans both domains and confers notable orange coloration to the protein (Fig. 1B). Upon illumination, the OCP structure dramatically changes: carotenoid slides into the NTD, the NTE dissociates from the CTD and domains become separated[15–19] (Fig. 1C, step 1), which is also accompanied by a spectral transition and hence a color change from orange ($\mathrm{OCP^O}$) to red ($\mathrm{OCP^R}$)[11] (Fig. 1B). During NPQ mediated by OCP,[11,20] such structural changes drive the binding of the NTD to the light-harvesting antennas, phycobilisomes (PB);[12] a similar effect is achieved by the individual NTD harboring carotenoid[17] but not by the CTD.[17,21] On binding to PB, OCP causes fluorescence quenching (Fig. 1C, step 2). OCP intercepts the energy flow from the PB core, dissipating it into heat and thereby preventing energy transfer to the photosynthetic reaction centers.[12] Orange $\mathrm{OCP^O}$ is generally associated with the resting state, which coincides with the obtained crystal structures (PDB codes: 3MG1, 4XB5), whereas red $\mathrm{OCP^R}$ is associated with the photoactivated, physiologically active form capable of PB quenching. Recently, the "orange" and "red" intermediates of the photoactivation process of OCP have been detected,[22,23] indicating that the simple association of the functional state of this protein with the color may be inadequate. This is further supported by the recent data on an OCP variant with the NTE trapped on the CTD by an engineered disulfide (termed OCPcc), which uncouples the ability of OCP to undergo reversible photoactivation from structural rearrangements essential to its photoprotective function.[24] While being indispensable for the detailed understanding of the photoprotection mechanism, unlike $\mathrm{OCP^O}$, the atomic structure of the full-length $\mathrm{OCP^R}$ remains elusive due to its dynamic and metastable nature that challenges structural studies. At the same time, the photoactivated OCP state can be mimicked by mutation of the conserved ketocarotenoid-coordinating residues in the CTD (namely W288A or W288A/Y201A) inducing domain separation, spectral shift and other signatures of the functional $\mathrm{OCP^R}$ state.[25]

Discovery of FRP and its function

Soon after revealing the role of OCP in the process of NPQ, the second main regulator was discovered.[13] After the deletion of one gene proximal to OCP, that is *slr1964*, PB fluorescence recovery after blue light-induced NPQ was significantly diminished.[13] The efficiency of fluorescence quenching was only slightly increased compared to the wild-type cells indicating that the *slr1964* gene product was not directly involved in the PB quenching formation. Due to its physiological effect, such a protein was named Fluorescence Recovery Protein (FRP).

The *in vitro* reconstitution study in *Synechocystis* sp. PCC 6803 (hereinafter, *Synechocystis*)[12] demonstrated that the minimum set consisting of PB, OCP and FRP is sufficient for the modeling of the photoprotection mechanism. The addition of FRP to the stably quenched PB–OCP complex resulted in PB fluorescence recovery (Fig. 1C, step 3-a), whereas the presence of comparable amounts of FRP prevented OCP from PB fluo-

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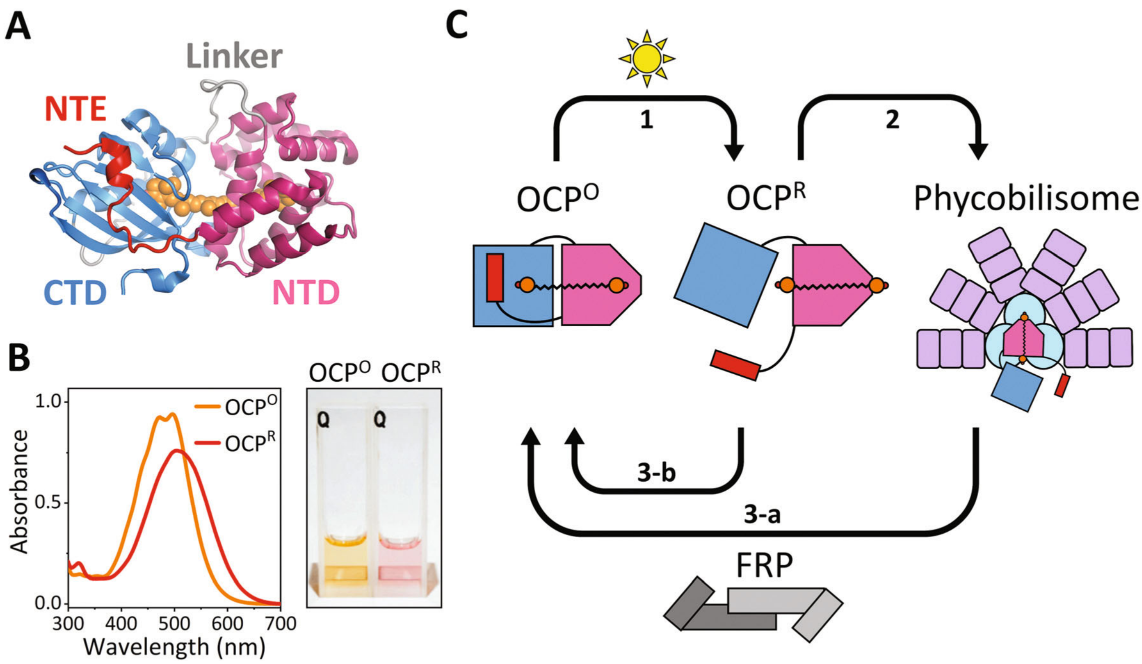

Fig. 1 Structure and function of OCP. (A) Crystal structure of OCP from *Synechocystis* sp. PCC 6803 (PDB code 4XB5). Carotenoid molecule (orange) is burrowed between the N-terminal (NTD, pink) and C-terminal (CTD, blue) domains. The domains are connected by a linker polypeptide (grey) and stabilized by the N-terminal extension (NTE, red) binding to the β-sheet surface of the CTD. (B) Spectral and color changes of OCP solution upon photoactivation. (Left) Upon blue-green light illumination, the OCP absorbance spectrum shifts to longer wavelengths, broadens and looses the fine structure. (Right) The significant spectral shift leads to the remarkable color change from orange to red. (C) A simplified model of the OCP-dependent photoprotection mechanism. OCP (same color coding) is photoactivated and undergoes dramatic structural transformations (step 1). After the carotenoid movement into the NTD and domain separation, OCP acquires the ability to bind to the PB core (*cyan* circles) and intercept the electronic excitation energy of PB pigments (step 2). FRP accelerates the detachment of OCP from the PB-OCP complex (step 3-a) and the conversion of OCP^R to OCP^O (step 3-b).

rescence quenching even under actinic illumination.[12,25] Although the blue light-triggered OCP^O–OCP^R transition and OCP^R binding to PB were slowly reversed in the dark, this process was significantly accelerated by the action of FRP[13] (Fig. 1C, step 3-b). FRP was proposed to deliver two distinct functionalities,[27] one being detachment of OCP from PB and the second being the acceleration of OCP inactivation by facilitated domain re-association (Fig. 1C, steps 3-a and 3-b). Basically, the main role of FRP is now recognized to consist of OCP down-regulation and control over the efficiency of OCP-mediated photoprotection.

FRP gene arrangement and expression

FRP genes are located downstream of OCP genes across cyanobacteria in two patterns: in tandem or with a β-carotene ketolase gene inserted in between.[13] In a few cyanobacterial genomes such as *Pleurocapsa* sp. PCC 7319, *Calothrix* sp. PCC 6303 and *Calothrix desertica* PCC 7102, two copies of FRP genes could be detected,[3] but whether both of the corresponding FRP gene products are functional or not is an open question. In the case of *Calothrix desertica* PCC 7102, there are two gene clusters containing OCP and FRP in its genome and both FRP sequences are relatively similar (52.1% protein sequence identity), likely indicating a duplication event. The tandem arrangement of OCP and FRP genes led to the hypothesis of their co-transcriptional regulation. It turned out that under the natural promoter, the transcript containing both complete genes was not detected and FRP mRNA was present even after the disruption of the OCP gene, indicating the existence of its own promoter.[13] However, after introduction of the strong psbA2 promoter before the OCP gene, the apparent operon transcript with both genes could be obtained, at least in principle.[13] The precise mechanism of transcriptional regulation of OCP and FRP is yet to be elucidated. Overexpression of FRP resulted in a significantly faster fluorescence recovery emphasizing the importance of a proper balance in the expression levels of OCP and FRP for the efficient regulation of the photoprotection mechanism.[13,26,27] Unfortunately, the FRP transcript level has not been properly evaluated so far, presumably due to its relatively low expression, whereas the transcription level of OCP was successfully determined.[28] According to immunoblot studies, a much higher OCP concentration over FRP is required for the induction of photoprotection in *Synechocystis*,[26] which is in agreement with the results of an *in silico* study modeling the processes of the photoprotection mechanism.[27] This apparently contradicts the global gene expression profiling of *Synechococcus* sp. PCC 7002 that presents the OCP and FRP mRNA quantity of the same order.[29]

An accurate evaluation of the protein and mRNA levels of OCP and FRP under different physiological conditions as well as a comprehensive description of OCP and FRP promoters warrant further detailed studies.

Structural organization of FRP

Optimization of heterologous protein expression and purification from *Escherichia coli* provided the fully functional water-soluble FRP suitable for characterization and structural analysis.[26,30] The atomic structure of FRP was determined by X-ray crystallography using the protein from *Synechocystis* (PDB code 4JDX)[30] or *Tolypothrix* sp. PCC 7601 (PDB code 5TZ0)[28] and the solution structures of FRP from *Synechocystis*, *Anabaena variabilis* ATCC 29413 and *Arthrospira maxima* CS-328 were studied by small-angle X-ray scattering.[31] FRP is an α-helical dimeric protein (Fig. 2A, left) consisting of four α-helices and one penultimate short 3_{10}-helix organized into two distinct parts which were termed "head" (cyan) and "stalk" (blue) (Fig. 2B). The stalk consists of α1 and α2 helices (Fig. 2A, right) providing a hydrophobic interface that stabilizes the dimeric assembly. The disturbance of this hydrophobic core by an engineered negative charge (*e.g.*, the L49E mutation) leads to an almost complete FRP monomerization.[32] The FRP dimer is also stabilized by the essential π-cation interaction between the side chains of R60 and W50 residues and also the H-bonds between D54 and R60 residues (Fig. 2B). The substitution of either of them leads to FRP dysfunction (Table 1), although the R60K mutant form appeared to preserve the dimeric assembly typical of the wild-type FRP crystal structure.[30] The rest of the FRP helices form the head domain

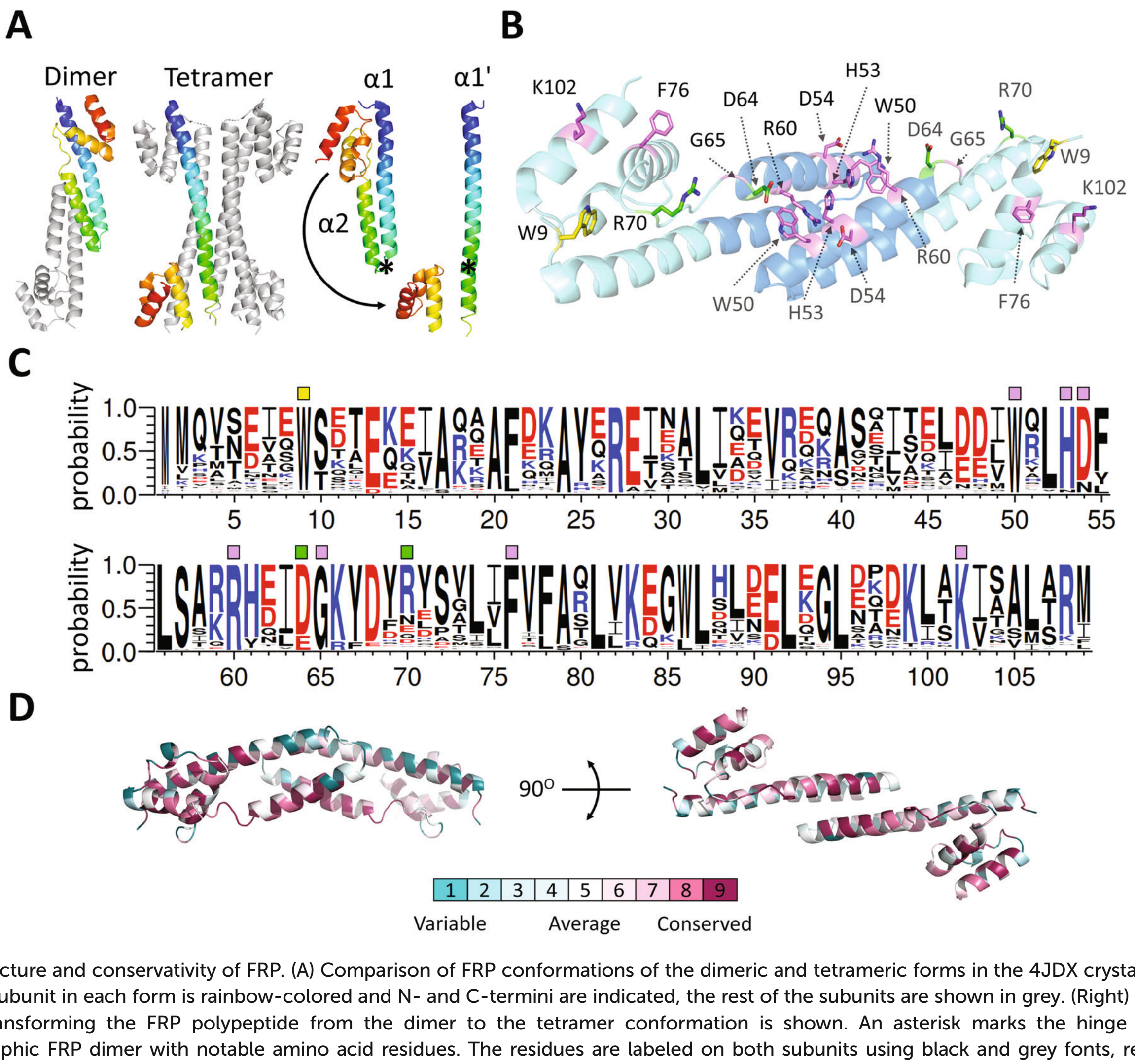

Fig. 2 Structure and conservativity of FRP. (A) Comparison of FRP conformations of the dimeric and tetrameric forms in the 4JDX crystal structure. (Left) One subunit in each form is rainbow-colored and N- and C-termini are indicated, the rest of the subunits are shown in grey. (Right) An imaginary turn transforming the FRP polypeptide from the dimer to the tetramer conformation is shown. An asterisk marks the hinge region. (B) Crystallographic FRP dimer with notable amino acid residues. The residues are labeled on both subunits using black and grey fonts, respectively. Distinct FRP subdomains, "head" and "stalk", and shown in cyan and blue, respectively. Side chains are shown by sticks colored according to the effect of a particular amino acid substitution on the functional activity of FRP: purple – reduced, green – enhanced, yellow – no effect. More details are summarized in Table 1. (C) Conservativity diagram built with WebLogo[44] based on the multiple sequence alignment of 88 unique cyanobacterial FRP sequences with amino acid residues important for functioning labeled by colored squares (same color coding as in panel B). (D) Conservativity mapping on the FRP crystal structure (PDB code 4JDX) using ConSurf,[45] the color coding scheme is presented.

Table 1 Notable amino acid substitutions in FRP and OCP affecting their interaction

Amino acid substitution[a]	Description of the reported effect	Methods
	FRP	
R60L	Reduced FRP-dependent acceleration of OCP R–O conversion	Abs. spectroscopy[30,34,35]
	Interaction between OCP and FRP is not abolished since the FRP mutant co-precipitated CTD-OCP	Co-immunoprecipitation[35]
	This amino acid substitution slightly slows down FRP-accelerated OCP detachment from PB	Fluorescence steady-state spectroscopy[35]
	The pattern of the FRP-CTD/FRP-CTD-NTD peaks detected by native mass-spectrometry is similar to that in the case of the wild-type FRP, indicating almost no effect on FRP binding	Native mass-spectrometry[34]
W50L|W50F|H53L D54L|D54E|R60K	Reduced FRP-dependent acceleration of OCP R–O conversion	Abs. spectroscopy[30]
W9L	No effect on the FRP-dependent acceleration of OCP R–O conversion	Abs. spectroscopy[30]
F76D|K102D	Reduced FRP-dependent acceleration of OCP R–O conversion	Abs. spectroscopy[34]
	Peaks of FRP-CTD and FRP-CTD-NTD complexes almost disappear, whereas peaks of dimeric FRP are present in native mass-spectrometry indicating a weakening of FRP binding	Native mass-spectrometry[34]
G65D	Reduced FRP dependent acceleration of OCP R–O conversion	Abs. spectroscopy[34]
	The pattern of the FRP-CTD/FRP-CTD-NTD peaks detected by native mass-spectrometry is similar to that in the case of the wild-type FRP, indicating almost no effect on FRP binding	Native mass-spectrometry[34]
D64A|R70D	Enhanced FRP-dependent acceleration of OCP R–O conversion	Abs. spectroscopy[34]
	The pattern of the FRP-CTD/FRP-CTD-NTD peaks detected by native mass-spectrometry is similar to that in the case of the wild-type FRP, indicating almost no effect on FRP binding	Native mass-spectrometry[34]
	OCP[b]	
D220K	Reduced FRP-dependent acceleration of OCP R–O conversion	Abs. spectroscopy[35]
	Almost no effect on OCP detachment from PB	Fluorescence steady-state spectroscopy[35]
N236K	Almost no effect on the FRP-dependent acceleration of OCP R–O conversion	Abs. spectroscopy[35]
K167A	No effect on the FRP-dependent acceleration of OCP R–O conversion	Abs. spectroscopy[40]

[a] Amino acid substitutions are grouped according to their effects on FRP function and methods of FRP binding/activity evaluation. [b] Note that we took into account only those amino acid substitutions in OCP which did not affect the OCP photocycle and, therefore, whose effects could be directly interpreted in terms of the FRP binding.

present in each FRP subunit, which is also important for the functioning of FRP (Fig. 2B). The described tertiary structure of FRP (Pfam: PF18032) has so far remained unique.

Of note is the fact that there is an alternative conformation of the FRP polypeptide chain, wherein α1 and α2 form one long α1′ helix (Fig. 2A, right). Moreover, such a conformation produces a distinct tetrameric crystallographic form (Fig. 2A, middle). The tetrameric FRP was not detected in the crystal structure of FRP from *Tolypothrix* sp. PCC 7601 (PDB ID 5TZ0)[28] and unlikely forms in an unconcentrated FRP solution,[25,30,33] but the difference in the polypeptide chain conformations observed in the 4JDX structure suggests probable structural dynamics of FRP that may be associated with its function. Indeed, a comparison of the two conformations reveals the possibility of bending of the long α1′ helix with a clear hinge region (Fig. 2A, right). Noteworthily, this rearrangement is also accompanied by some local structural unfolding. On top of that, an intersubunit angular shift in FRP dimers suggested by comparison of the crystallographic dimers of FRP from *Synechocystis* and *Tolypotrix*[28] and detected by small-angle X-ray scattering (SAXS) shows the possibility of sliding, angular movements of FRP subunits along the hydrophobic dimer interface.[31] Conserved amino acid residues (Fig. 2C) are predominantly located only on one side of the FRP dimer and form distinct patches in the stalk and head regions (Fig. 2D). The remarkable distribution of such amino acid residues led to the hypothesis of their involvement in FRP activity and its verification through numerous mutational studies[30,32,34,35]

(Fig. 2C). Mutation of many conserved amino acids negatively affected the FRP function; however, the precise molecular mechanism of the OCP–FRP interaction remains far from being completely understood.

Features of the OCP–FRP interaction

The physiological activity of FRP reversed the OCP-dependent PB quenching, therefore the direct interaction between OCP and FRP was proposed. Indeed, co-immunoprecipitation of OCP and FRP using anti-FRP antibodies verified this interaction and demonstrated that it increased upon OCP photoactivation.[13] Later on, the direct interaction between FRP and OCP^O or the photoactivation mimicking OCP mutant W288A was measured quantitatively using analytical size-exclusion chromatography (SEC) and native gel-electrophoresis.[25] It was demonstrated that the FRP interaction with OCP^R (or with the W288A mutant) was at least 10 times stronger than that with OCP^O.[25] The interaction of FRP with the apoform of OCP was also shown at an affinity comparable to that to OCP^R or its functional analog W288A.[36] Together this indicated that FRP specifically recognizes OCP forms with the separated domains while binding with low affinity to the resting OCP form. Clearly, this is in line with the assumption that the FRP interaction with OCP^R leads to re-association of OCP domains, which, in turn, yields OCP^O and causes an FRP release from the complex, resembling the action of an enzyme.

However, it turned out that, in contrast to enzymes, FRP only slightly lowers the activation energy of the R–O transition.[25] The thermodynamic effect of FRP on OCP R–O conversion predominantly affects the pre-exponential factor of the Arrhenius equation, which may be considered as lowering of the number of attempts required for OCP domains to find each other in a proper orientation and assemble in the resting state.[25] In line with this, the OCP domain bridging (or scaffolding) activity of FRP was proposed,[21,25,37] which questioned the location of the FRP binding site(s) on OCP.

There is compelling evidence that the main site for FRP binding is located on the CTD of OCP. The FRP interaction with the CTD of OCP was shown by co-immunoprecipitation,[30,35] SEC[21] and native mass-spectrometry.[34] It was directly shown that the deletion of the NTE, which normally attaches to the β-sheet surface of the CTD within OCP^O, unleashes tight FRP binding to such a deletion OCP mutant (named ΔNTE-OCP) even without its photoactivation.[37] Importantly, the deletion of NTE destabilized OCP and, upon its expression in *E. coli* cells producing ketocarotenoids,[38] yielded a mixture of OCP^O- and OCP^R-like species differing by color, absorbance spectrum and protein compactness.[37] Binding of FRP leads to the so-called "oranging" (*i.e.* increased proportion of OCP^O) of the ΔNTE-OCP absorbance spectrum associated with the domain bridging activity of FRP.[25,37] The tight photoactivation-independent interaction of FRP with ΔNTE-OCP strongly suggested that the binding sites of FRP and NTE on the surface of the CTD overlap (if not coincide).[32,37] Remarkably, this was recently additionally confirmed by the orange OCP mutant with the disulfide-trapped NTE (OCPcc), which was completely unable to interact with FRP unless the engineered disulfide bridge between the NTE and CTD was uninstalled by reduction.[24] The absence of any signs of interaction in the case of the trapped OCPcc and FRP also indicated that the low-affinity binding of FRP to OCP^O is likely explained by FRP partially outcompeting the NTE from its binding site on the CTD and by this means disproved the alternative binding modes within the FRP/OCP^O complexes proposed earlier.[25,34]

Given the tight association of FRP with the CTD of OCP, it likely serves the initial docking point, whereas the tentative domain bridging activity of FRP implies the existence of some transient secondary interaction(s) between FRP and the NTD or the linker region. FRP binding to the individual NTD could not be detected by analytical SEC,[21] but was demonstrated by native mass-spectrometry confirming the existence of the weak NTD/FRP binding site.[34] Of note is that the NTD/FRP complex was detected only after collision-induced dissociation of the higher-order CTD/NTD/FRP complex, suggesting the likelihood of FRP "activation" by the initial binding event for the following secondary interaction with the NTD, in line with the inability to observe the direct FRP binding to the NTD.[21]

The normally dimeric FRP shows a well-documented tendency to monomerize upon binding to OCP,[25,31,32,34,36,37] indicating that OCP binding likely induces significant structural rearrangements in its partner. Indeed, solvent accessibility assays showed that both the FRP head and stalk may undergo dramatic transformations upon interaction.[34,39] Monomerization of FRP seems to represent a distinct stage of the dynamic OCP–FRP interaction during FRP functioning.[25,31,34,37] Monomerization of FRP seems to occur only after the FRP dimer is bound to OCP, as the engineered FRP monomer was poorly efficient, in contrast to the engineered FRP dimer that was fully active in terms of binding and functional activity.[32] In practice, OCP–FRP complexes with both 1 : 1 and 1 : 2 stoichiometries could be observed, which seems to depend on the protein concentration ratio and on the FRP homolog.[31] Interestingly, dimeric FRP from *Arthrospira maxima* CS-328 preferentially forms a 1 : 1 OCP–FRP complex,[31] providing a convenient model for studying such a multistep process, which is most likely composed of several intermediary steps corresponding to complexes with different stoichiometries and having protein partners in different conformations.

Effect of sequence variations in FRP and OCP on their physical interaction

In the absence of detailed structural information on the OCP/FRP binding interfaces, some knowledge may be gained from mutational studies and sequence variations. We have grouped all amino acid substitutions in FRP and OCP that are known to affect their interaction (Table 1, Fig. 2B). Clearly, the effects of these substitutions are not evenly characterized, and most of the conclusions with respect to the interaction were drawn from functional tests only (*i.e.*, acceleration of OCP R–O conversion in the presence of a particular sequence variant of FRP), without a relevant assessment of the physical interaction between the proteins.

No study describing FRP residues responsible for the interaction with the OCP NTD (or *vice versa*) is available so far. At the same time, many amino acid substitutions in OCP disrupting its interaction with FRP are described and located on the CTD,[35] apparently supporting the primary role of the CTD in FRP recruitment. Nonetheless, only two of them, that is D220K and N236K, while reducing the FRP-dependent acceleration of R–O conversion, do not disturb the OCP photocycle itself (Table 1). One can argue that the convenient assessment of the OCP–FRP interaction through the FRP effect on OCP photoconversion provides reasonable interpretations only if supported by additional controls and distinguishing between the effects on the OCP photocycle, the physical OCP–FRP interaction, and, finally, the functioning of this physiological mechanism in a given OCP/FRP combination. Indeed, amino acid substitutions in OCP often alter its photocycle, which may prevent the FRP-dependent acceleration of R–O conversion without an immediate disturbance of OCP–FRP binding. For example, the stability of OCP^O was substantially compromised in the F299R mutant,[35] and the observed absence of the FRP action on this mutant may be interpreted either as weakening of the OCP–FRP interaction or as the inability of FRP to overcome the destabilizing effect of F299R substitution itself. On the other hand, even the inhibition of the FRP action in the case of some FRP mutants does not necessarily mean the lack of the OCP–FRP interaction, as is clearly the case for the FRP (R60L) mutant[34,35] (Table 1 and Fig. 2B). This may be better explained by the putative existence of more than one binding interface stabilizing the OCP/FRP complex, when FRP binding would not guarantee the functional activity providing OCP domain re-association. Alternatively, point substitutions in the dimer interface could destabilize the structure of FRP leading to its monomerization, partial unfolding and thus the inability to bind OCP, as was shown for the engineered FRP monomer with the L49E substitution.[32] Analysis of the OCP–FRP interaction only through the effects on OCP photoconversion thus inevitably leads to the ambiguity of interpretations. Unfortunately, most of the studied amino acid substitutions in both proteins were characterized in such a way (Table 1).

The recent bioinformatic analysis revealed the existence of two additional paralogous OCP subfamilies (OCP2 and OCPX).[28] These paralogs often co-exist in one genome with the "canonical" OCP1 (here "canonical" has the only meaning that it is better studied). For example, *Tolypothrix* sp. PCC 7601 possesses both OCP1 and OCP2 genes, however the expression level of OCP2 reaches that of OCP1 only under high light conditions.[28] This led to the hypothesis that OCP2 homologs play a role in the acclimation of cyanobacteria. Although conserving many positions of the OCP1 consensus sequence, paralogous OCP2 and OCPX show a faster back-conversion to the resting orange state and a weaker interaction with PB upon photoactivation compared to OCP1.[28,40] Intriguingly, R–O conversion of OCP2 and OCPX appears to be not influenced by FRP[28,40] but no data are available on the physical binding between FRP and these OCP paralogs, which remains, therefore, not ruled out. In support of the opposite hypothesis, there are only a few notable substitutions in OCP2 and OCPX CTDs that, according to the current knowledge, may actually interfere with the FRP binding (R229E, D262G).[40] If the binding still occurs, one cannot exclude that FRP plays a role of detaching OCP2 and OCPX from their complexes with PB, as was shown in the case of the FRP(R60L) mutant and OCP1.[35]

Models describing the OCP–FRP interaction

Unfortunately, exhaustive attempts to crystallize the OCP–FRP complex so far have been unsuccessful. Application of other structural methods and special techniques detecting the footprints of protein–protein interactions yielded several models of OCP–FRP interaction. All of them agree on the FRP conservative head binding to the CTD of OCP and the importance of the NTE as a structural constraint. However, they are completely different in details. We divided the current models of OCP–FRP binding into two principal groups: FRP binding near the interdomain cavity (type I) and FRP binding on top of both OCP domains (type II). One of the most debatable points in both types of models is where the FRP head is located during the interaction. Based on independent methods, models place the FRP head (i) in place of the NTE on the CTD[32] (Fig. 3, red arrow), (ii) towards the linker, perpendicular to the common domain axis[34] (Fig. 3, dark green arrow) or (iii) opposite to the linker[39] (Fig. 3, dashed light green arrow). First, we will describe the "binding between the domains" (type I) model and then "binding on top of the domains" (type II) model.

Chemical cross-linking of the OCP–FRP complex and consequent peptide analysis suggested FRP binding in the interdomain cavity in OCP^R close to the linker region (Fig. 3, dark green arrow). Solvent accessibility assay demonstrated significant dynamics of FRP head residues upon interaction with OCP^R, suggesting the local unfolding of FRP, in line with the FRP "activation" hypothesis and conformational dynamics of the FRP polypeptide chain discussed above. According to the model, such unfolding may provoke monomerization and redistribution of charged amino acid residues permitting FRP monomer binding in the interdomain cavity[34] (1 : 1 OCP–FRP complex). Bound in the cavity FRP destabilizes the OCP^R state, thus accelerating R–O conversion. Further OCP domain closure displaces FRP from the cavity in the final step of their interaction. Intriguingly, chemical cross-linking coupled to mass-spectrometry[34] could detect even the weak OCP^O–FRP interaction shown previously by SEC.[25] OCP^O–FRP cross-linking was significantly different from the OCP^R–FRP counterpart as only a single OCP(K167)–FRP(M1) cross-link could be obtained in the former case, indicating FRP head binding

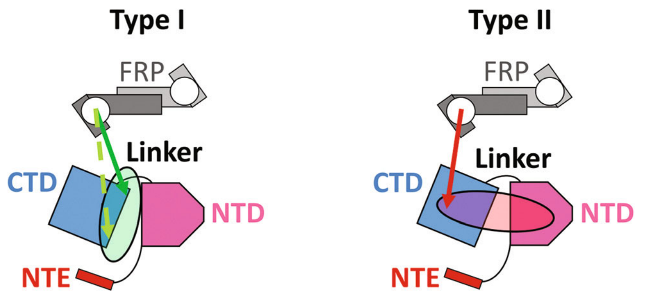

Fig. 3 OCP–FRP binding models (carotenoid is omitted for clarity). The orientation of the FRP head in the models is marked by a corresponding colored arrow. Both model types propose the FRP head (white circle) as the main site of FRP binding on OCP, which is consistent with different OCP–FRP complex stoichiometries.[25,31,32,34,37] In the first type of models, the FRP head occupies the interdomain cavity that becomes exposed due to OCP domain separation (green oval). Depending on the methods used, the orientation of the FRP head relative to the CTD diverges (green arrows). Of note is that FRP binding close to the linker region (dark green arrow) suggests the molecular basis of the FRP action through binding to both the CTD and the linker.[34] A significant decrease in the solvent accessibility of the CTD residues opposite to the linker (dashed light green arrow) during the OCP–FRP interaction suggests that the FRP head masks them.[39] In the second type model, the FRP head occupies the NTE binding site (red arrow) and the rest part of the FRP dimer (or the monomer, depending on the intermediate) lies on top of the OCP domains (red circle), facilitating their re-association. Such a topology is principally compatible with the binding of two OCP molecules to the two FRP head domains simultaneously (and is supported by disulfide trapping and chemical crosslinking[32]), whereas FRP monomerization, often observed occurring to a different degree,[25,31,32,34,37] may be facilitated by putative binding of the additional OCP molecule to the OCP–FRP (1 : 2) complex through the vacant FRP head.

close to the OCP linker region without intercalating into the OCP interdomain cavity.[34]

Alternatively, X-ray radiolytic labeling shows significant masking of the FRP head residues (71–83) and the CTD residues (M284, P276, W277, F278) located on the CTD side opposite to the interdomain linker[39] (Fig. 3, dashed light green arrow). The corresponding model proposed recently[39] includes the bend of the α1-helix of FRP and its binding to the CTD in place of the NTE, which, however, is not based on experimental data. Such a model also implicates the monomerization of FRP, as the α1-helix is directly involved in the formation of the dimeric assembly of FRP. One of the most controversial points in this study is an almost indistinguishable effect on the solvent accessibility of both protein partners upon OCP^R–FRP and OCP^O–FRP interaction.[39] This raises the question of whether it was indeed the OCP^R–FRP interaction that was detected and analyzed, or whether the obtained results refer to the weak OCP^O–FRP complex. Unfortunately, no orthogonal approach was used to support the findings. Anyway, X-ray radiolytic labeling data remain useful and may be adopted with care in other OCP–FRP interaction models. For example, X-ray radiolytic labeling indicates that FRP may displace the NTE even in OCP^O, which may be interpreted as the binding of the FRP head on the CTD and may be validated through fixation of the NTE on the CTD. This has been directly confirmed recently by the use of the OCPcc mutant with the engineered disulfide bridge trapping the NTE.[24]

The second type model, based on the integrative approach using SAXS and complementary biochemical techniques, locates the FRP molecule upon both OCP domains nearly parallel to the common domain axis, with the FRP head anchored on the CTD in place of the NTE (Fig. 3, red arrow). This is arguably the most detailed model of the OCP–FRP binding reported to date that is compatible with the conformational change in FRP and also accounts for FRP monomerization and the existence of several intermediates corresponding to OCP–FRP complexes with different stoichiometries. The key component necessary for the elucidation of the OCP–FRP complex architecture was the stabilization of the OCP–FRP (1 : 2) complex, which may be viewed as the intermediate state after the recruitment of FRP to the CTD of OCP, by using an engineered FRP dimer unable to dissociate due to the introduced disulfide bridges (FRPcc).[32] FRP with the retained ability to dissociate upon OCP binding inevitably led to a mixture of OCP–FRP complexes with different stoichiometries (1 : 1 and 1 : 2) as well as OCP molecules at different states of the OCP photocycle. Such multi-level heterogeneity seriously complicates structural studies in general and may be the main cause of the continuous unsuccessful attempts to crystallize the OCP–FRP complexes.

Unexpectedly, the cross-linked FRPcc dimer was capable to accelerate R–O conversion, indicating that monomerization is not decisive for the functional activity of FRP. Cross-linking of the OCP–FRPcc mixture by glutaraldehyde led to the formation of complexes with the previously unseen 2 : 2 stoichiometry (yet formation of higher-order species could not be observed), showing the principal possibility of the vacant FRP head in the 1 : 2 OCP–FRP complex to bind an extra OCP.[32] The OCP–FRP 2 : 2 complex was not detected in solution presumably because of the steric clash between OCP molecules in such a protein ensemble, which is overridden by cross-linking. Apparently, FRP monomerization occurs as a by-product of the OCP–FRP interaction either due to FRP "activation" upon the primary binding event or after binding of the second OCP molecule to the OCP–FRP (1 : 2) complex. FRP monomerization was proposed to improve the FRP efficiency upon exceeding OCP concentrations, which is estimated to virtually always be the case in the cellular context.[32]

The corresponding OCP–FRP complex topology was supported by disulfide trapping of the FRP head residue (K102C) with the CTD residue (F299C) that is normally masked by the NTE in OCP^O.[32] Both these residues are conserved in FRP, and their mutation to Cys residues used for trapping appreciably decreased the OCP–FRP binding affinity and therefore the efficiency of cross-linking, indirectly supporting that they are located close to the protein–protein interface. The overall distribution of conservative residues only on one side of FRP was also very consistent with the proposed orientation of the FRP molecule within the studied 1 : 2 OCP–FRP complex. Moreover, the proposed topology was consistent with the complementary surface distribution of charges in interacting proteins.[32] Of note is that the represented OCP–FRP binding model does not inform us about the conformational changes likely taking place upon the OCP–FRP interaction, although it does not exclude it. Likewise, it provides just a single snapshot in the multistage dynamic process, in the course of which one can expect dislocation of the proteins relative to each other, accompanied by serious structural rearrangements. The additional techniques are, therefore, required for the elucidation of the full molecular mechanism of the FRP action. We emphasize that distinct stages of the FRP action are still poorly defined and understood. Significant structural changes of FRP during the cooperative OCP–FRP cycle may reconcile contradicting models, but require verification through the stabilization of different FRP states.

Phylogenetic tree reveals several clades of non-cyanobacterial FRP-like proteins

Many efforts from different laboratories helped in outlining the photoprotection mechanism regulated by the OCP–FRP interplay in *Synechocystis*; however, little is known about its universality across cyanobacteria.

Pairwise alignment of available FRP protein sequences shows that their identity level sharply drops to 40–50% outside of the seed FRP genera. At first glance, this not only indicates the relatively long evolutionary history of FRP but questions whether FRP is similarly functional in different cyanobacteria (that is, the "universality" hypothesis). It was shown that low-homology FRP representatives from *Arthrospira maxima* CS-328 and *Anabaena variabilis* ATCC 29413 may accelerate R–O conversion and detachment from PB of OCP from *Synechocystis* nearly as efficient as *Synechocystis* FRP does, despite only 37.6% protein sequence identity among these three FRP sequences.[31] This partially mitigated the severity of the issue and implied that the sequence positions remaining evolutionarily "cold" (Fig. 2D) may be sufficient for the activity of this protein.

In order to update the phylogenetic tree constructed earlier,[31] we searched for all available FRP sequences from different databases. Curiously, the BLAST search against the comprehensive nucleotide collection which consists of GenBank, EMBL, DDBJ, PDB, RefSeq and WGS sequences returns FRP-like hits even in non-cyanobacteria: Alfa-, Beta-, Gamma- proteobacteria, Acidobacteria, Chlorobi and even Euryarchaeota.

We constructed a phylogenetic tree (Fig. 4) based on available FRP-like protein sequences 101–117 residues long from all bacterial sources. Unexpectedly, almost all non-cyanobacterial FRP (see ESI Table 1† for the annotated list of their sequences) formed several clades in a large cluster being distinctly different from cyanobacterial FRP sequences, with only one exception representing a non-cyanobacterial FRP gene which is rather similar to FRP from *Synechococcus* genera. This gene was most likely horizontally transferred, as in *Alteromonadaceae* bacterium (Fig. 4, top left) the whole gene cluster consisting of OCP, carotene ketolase and FRP typical of cyanobacteria could be detected. The rest of the FRP-like genes from non-cyanobacteria are probably distant homologs with unknown functions. The existence of non-cyanobacterial FRP-like genes is striking as FRP is currently considered as an OCP partner only, which obviously cannot be the case in bacteria lacking OCP and PB. The majority of bacteria that possess FRP homologs (the suggested name is FRPH, similar to CTDH, the homologs of the C-terminal domain[41]) are chemoheterotrophic Proteobacteria with only one notable photoautotrophic exception – *Rhodoferax* sp. (Betaproteobacteria class, purple non-sulfur bacteria). FRPHs from different bacterial clades are completely uncharacterized, orphan proteins and this could be a fruitful area of research in the future.

From the tree in Fig. 4, we can also see that the diversity of even cyanobacterial FRP is far from being completely characterized (see also ESI Table 2† for the annotated list of their sequences). There are many extremely low-identity homologs (down to 31.4% compared to the protein sequence of FRP from *Synechocystis*) in *Synechococcus* genera, where both FRP and OCP are present. Characterization of such homologs is not only necessary to cover the FRP diversity and verify the FRP universality hypothesis, but also for the elucidation of the whole process of the FRP action. As was shown for FRP from *Arthrospira maxima* CS-328, FRP may form stable 1 : 1 complexes with OCP; therefore, comparative analysis of distant FRP with respect to the OCP binding may provide us with deeper insights into the molecular mechanism of FRP action. Among perspective FRP for comparative analysis is the FRP homolog from *Crocosphaera watsonii* where OCP is absent (Fig. 4, bottom right). Most probably, the OCP gene in this species was lost, as the closely related *Cyanothece* species[42] possesses both OCP and FRP genes.[28] *Crocosphaera watsonii* cyanobacterium raises the question of whether FRP can have other functional roles beyond the interaction with OCP and possibly even beyond photoprotection.

Homology modeling of non-cyanobacterial FRPHs suggests divergence of functions

The existence of numerous FRP-like sequences (FRPHs) is intriguing, which prompted us to predict the structure of their several representatives using homology modeling (Fig. 5). Unexpectedly, predicted by i-TASSER[43] without using any template secondary and tertiary structures, 3D-structural models of archaeal and bacterial FRPHs appear to be very similar to FRP from *Synechocystis* despite the low protein sequence identity level (25–40%). Notwithstanding the similarities of the overall fold and the logics of the tentative FRPH dimer architectures, there are some notable substitutions in such FRP-like proteins (*Synechocystis* FRP numbering): W50G, G65E, G65R, G65Q, F76W, F76I, K102E, K102A, K102S and A105E. These alternative amino acid residues occupying positions that are relevant for *Synechocystis* FRP function, question the preservation of the functionality of the corresponding FRPHs with respect to the "canonical" OCP and instead suggest different functional roles in the organisms lacking OCP.

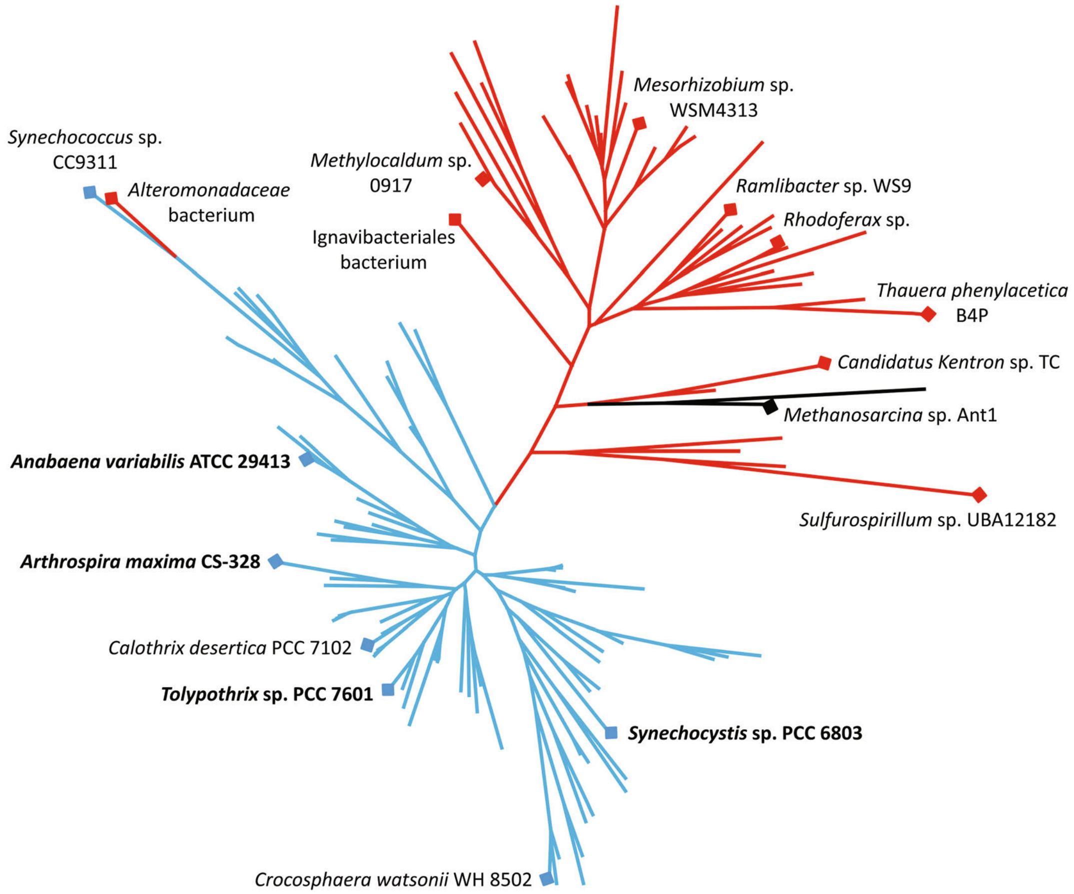

Fig. 4 Phylogenetic tree of 137 FRP-like sequences. Characterized FRP proteins from cyanobacteria are in bold. Representatives of non-cyanobacterial FRPH used for structural prediction alongside with the cyanobacterial FRP representatives mentioned above are also shown. Cyanobacterial, archaeal and other non-cyanobacterial FRP are colored with blue, black and red, respectively. The evolutionary history was inferred by using the Maximum Likelihood method and JTT matrix-based model in MEGA X.[46] The tree is drawn to scale, with branch lengths proportional to the number of substitutions per site. All positions containing gaps and missing data were eliminated. The topology of the presented tree can be found in ESI 3.† See also ESI Tables 1 and 2† for the annotated list of non-cyanobacterial and cyanobacterial FRPH sequences, respectively.

Comparative analysis of distant homologs may shed light on the amino acid residues that are essential for protein functioning as these residues should stay the same. Revealed by such analysis, diverged residues may specialize in protein–protein interactions between an FRPH and its partners. As mentioned above, there are several notable positions in non-cyanobacterial FRPHs located both in the head and the stalk. So far only substitutions of W50, G65, F76 and K102 residues were tested (W50L, W50F, G65D, F76D and K102D)[30,34] and it was clearly shown that such amino acid substitutions lead to significant preclusion of the FRP action on OCP. Further analysis and experimental characterization are necessary to clarify this apparent paradox. Curiously, the existence of FRP-like sequences in a "few bacterial genomes such as that of *Mesorhizobium ciceri*" was reported 7 years ago,[3] but no experimental work on them has been done since then. Numerous FRPHs with the fold reminiscent of FRP indicate the existence of yet unknown functions of this protein in bacteria and thus open new horizons for future research.

Perspectives

Over the decade since the discovery of FRP, the understanding of this protein has been significantly advanced and several tentative models have been put forward. However, the full mechanism of OCP–FRP binding could probably be unambiguously formulated only once the high-resolution structural data on the OCP–FRP complexes reflecting different stages of their molecular tango will become available. The evidence of multiple stages of the whole process implies that complementary biophysical studies are required for the evaluation of FRP dynamics. Among the striking events during the OCP–FRP interaction are the conformational change and monomerization of FRP that reflect the complexity of the FRP-dependent R–O acceleration mechanism. We consider obtaining the structure of the 1 : 1 OCP–FRP complex, an important intermediate stage of the FRP action, a missing element of the puzzle. The role of the well-documented FRP monomerization remains elusive and, in principle, may be a by-product of conformational rearrangements within FRP. The revealed disconnection

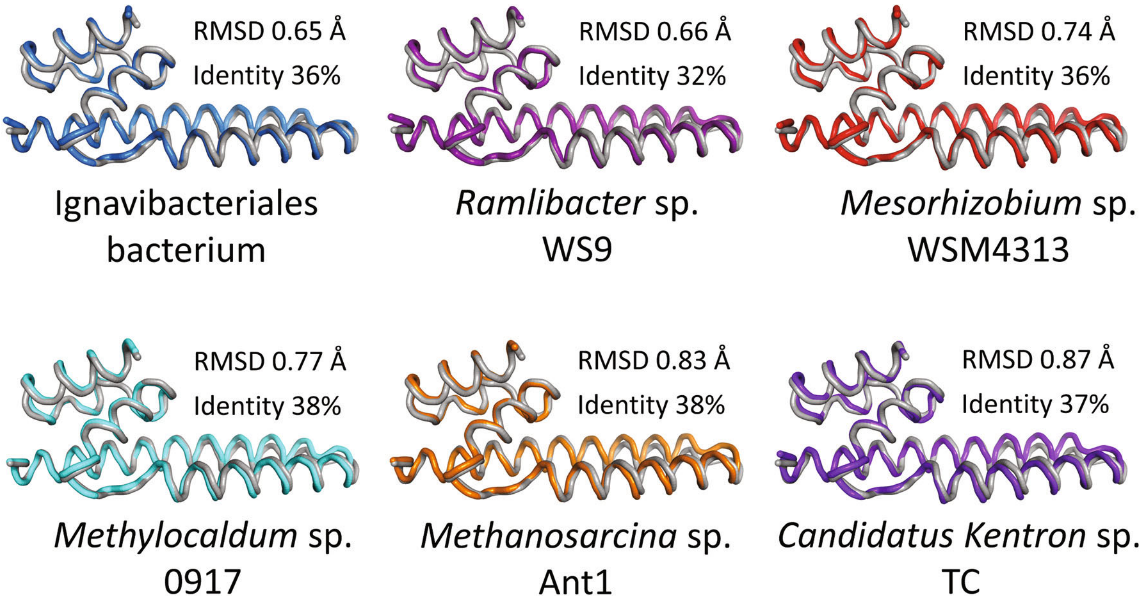

Fig. 5 Comparison of the FRP structure from *Tolypothrix* sp. PCC 7601 (grey ribbon) with the predicted FRPHs from non-cyanobacteria (ribbons labeled with different colors). Only one monomer is shown in each case for clarity. The Cα root mean square deviation (RMSD) calculated using i-TASSER[43] and the corresponding sequence identities determined upon pairwise alignment using Clustal Omega[47] are shown. "Blind" predictions of secondary and tertiary structures were made by i-TASSER[43] without using any templates. The best models were chosen for presentation and RMSD calculations. Images and structural overlays were done using PyMol.

between FRP binding and activity outlined the complexity of such a process and allows proposing an FRP "conformational cycle". This paradigm requires a revisiting, comprehensive analysis of the function, binding and dynamics of the FRP molecule of each previous and further FRP mutant forms. The starting point of OCP–FRP binding is likely the FRP dimer binding to the CTD of OCP, which initiates the sequence of events culminating in the re-association of OCP domains. The concrete effect of FRP binding to the CTD on the FRP structure within the complex may shed light on the early stages of the OCP–FRP interaction according to the FRP "activation" hypothesis.

Most likely, we stand at the beginning of the true understanding of the FRP mechanism, with many questions waiting for their answers: how does FRP provide its bridging activity and does it involve binding to the linker region or NTD? What is the role of the discovered FRPHs in different groups of bacteria, where its function is completely uncharacterized and likely lies beyond the light-harvesting regulation? Does cyanobacterial FRP play other roles apart from that in the OCP-mediated mechanism?

Conflicts of interest

The authors declare that they have no conflicts of interest.

Acknowledgements

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References

1 D. Kirilovsky and C. A. Kerfeld, Cyanobacterial photoprotection by the orange carotenoid protein, *Nat. Plants*, 2016, **2**, 16180.
2 D. Kirilovsky and C. A. Kerfeld, The orange carotenoid protein in photoprotection of photosystem II in cyanobacteria, *Biochim. Biophys. Acta, Bioenerg.*, 2012, **1817**, 158–166.
3 D. Kirilovsky and C. A. Kerfeld, The Orange Carotenoid Protein: a blue-green light photoactive protein., *Photochem. Photobiol. Sci.*, 2013, **12**, 1135–1143.
4 S. Bailey and A. Grossman, Photoprotection in cyanobacteria: Regulation of light harvesting, *Photochem. Photobiol.*, 2008, **84**, 1410–1420.
5 M. A. Dominguez-Martin and C. A. Kerfeld, Engineering the orange carotenoid protein for applications in synthetic biology, *Curr. Opin. Struct. Biol.*, 2019, **57**, 110–117.
6 H. Bao, M. R. Melnicki and C. A. Kerfeld, Structure and functions of Orange Carotenoid Protein homologs in cyanobacteria, *Curr. Opin. Plant Biol.*, 2017, **37**, 1–9.
7 K. K. Niyogi and T. B. Truong, Evolution of flexible non-photochemical quenching mechanisms that regulate light harvesting in oxygenic photosynthesis, *Curr. Opin. Plant Biol.*, 2013, **16**, 307–314.
8 N. N. Sluchanko, Y. B. Slonimskiy and E. G. Maksimov, Features of Protein – Protein Interactions in the

Cyanobacterial Photoprotection Mechanism, *Biochemistry*, 2017, **82**, 1592–1614.

9 A. Sedoud, R. Lopez-Igual, A. Ur Rehman, A. Wilson, F. Perreau, C. Boulay, I. Vass, A. Krieger-Liszkay and D. Kirilovsky, The Cyanobacterial Photoactive Orange Carotenoid Protein Is an Excellent Singlet Oxygen Quencher, *Plant Cell*, 2014, **26**, 1781–1791.

10 M. Watanabe and M. Ikeuchi, Phycobilisome: Architecture of a light-harvesting supercomplex, *Photosynth. Res.*, 2013, **116**, 265–276.

11 A. Wilson, C. Punginelli, A. Gall, C. Bonetti, M. Alexandre, J.-M. Routaboul, C. A. Kerfeld, R. van Grondelle, B. Robert, J. T. M. Kennis and D. Kirilovsky, A photoactive carotenoid protein acting as light intensity sensor., *Proc. Natl. Acad. Sci. U. S. A.*, 2008, **105**, 12075–12080.

12 M. Gwizdala, A. Wilson and D. Kirilovsky, In vitro reconstitution of the cyanobacterial photoprotective mechanism mediated by the Orange Carotenoid Protein in Synechocystis PCC 6803., *Plant Cell*, 2011, **23**, 2631–2643.

13 C. Boulay, A. Wilson, S. D'Haene and D. Kirilovsky, Identification of a protein required for recovery of full antenna capacity in OCP-related photoprotective mechanism in cyanobacteria., *Proc. Natl. Acad. Sci. U. S. A.*, 2010, **107**, 11620–11625.

14 C. A. Kerfeld, M. R. Sawaya, V. Brahmandam, D. Cascio, K. K. Ho, C. C. Trevithick-Sutton, D. W. Krogmann and T. O. Yeates, The crystal structure of a cyanobacterial water-soluble carotenoid binding protein, *Structure*, 2003, **11**, 55–65.

15 S. Gupta, M. Guttman, R. L. Leverenz, K. Zhumadilova, E. G. Pawlowski, C. J. Petzold, K. K. Lee, C. Y. Ralston and C. A. Kerfeld, Local and global structural drivers for the photoactivation of the orange carotenoid protein., *Proc. Natl. Acad. Sci. U. S. A.*, 2015, **112**, E5567–E5574.

16 R. L. Leverenz, M. Sutter, A. Wilson, S. Gupta, A. Thurotte, C. Bourcier de Carbon, C. J. Petzold, C. Ralston, F. Perreau, D. Kirilovsky and C. A. Kerfeld, A 12 A carotenoid translocation in a photoswitch associated with cyanobacterial photoprotection, *Science*, 2015, **348**, 1463–1466.

17 R. L. Leverenz, D. Jallet, M.-D. Li, R. A. Mathies, D. Kirilovsky and C. A. Kerfeld, Structural and Functional Modularity of the Orange Carotenoid Protein: Distinct Roles for the N- and C-Terminal Domains in Cyanobacterial Photoprotection, *Plant Cell*, 2014, **26**, 426–437.

18 E. G. Maksimov, N. N. Sluchanko, K. S. Mironov, E. A. Shirshin, K. E. Klementiev, G. V. Tsoraev, M. Moldenhauer, T. Friedrich, D. A. Los, S. I. Allakhverdiev, V. Z. Paschenko and A. B. Rubin, Fluorescent Labeling Preserving OCP Photoactivity Reveals Its Reorganization during the Photocycle, *Biophys. J.*, 2017, **112**, 46–56.

19 H. Liu, H. Zhang, J. D. King, N. R. Wolf, M. Prado, M. L. Gross and R. E. Blankenship, Mass spectrometry footprinting reveals the structural rearrangements of cyanobacterial orange carotenoid protein upon light activation, *Biochim. Biophys. Acta, Bioenerg.*, 2014, **1837**, 1955–1963.

20 A. Wilson, G. Ajlani, J.-M. Verbavatz, I. Vass, C. A. Kerfeld and D. Kirilovsky, A soluble carotenoid protein involved in phycobilisome-related energy dissipation in cyanobacteria., *Plant Cell*, 2006, **18**, 992–1007.

21 M. Moldenhauer, N. N. Sluchanko, D. Buhrke, D. V. Zlenko, N. N. Tavraz, F.-J. J. Schmitt, P. Hildebrandt, E. G. Maksimov and T. Friedrich, Assembly of photoactive orange carotenoid protein from its domains unravels a carotenoid shuttle mechanism, *Photosynth. Res.*, 2017, **133**, 327–341.

22 E. G. Maksimov, N. N. Sluchanko, Y. B. Slonimskiy, E. A. Slutskaya, A. V. Stepanov, A. M. Argentova-Stevens, E. A. Shirshin, G. V. Tsoraev, K. E. Klementiev, O. V. Slatinskaya, E. P. Lukashev, T. Friedrich, V. Z. Paschenko and A. B. Rubin, The photocycle of orange carotenoid protein conceals distinct intermediates and asynchronous changes in the carotenoid and protein components, *Sci. Rep.*, 2017, **7**, 15548.

23 P. E. Konold, I. H. M. van Stokkum and F. Muzzopappa, Photoactivation Mechanism, Timing of Protein Secondary Structure Dynamics and Carotenoid Translocation in the Orange Carotenoid Protein, *J. Am. Chem. Soc.*, 2019, **141**, 520–530.

24 Y. B. Slonimskiy, E. G. Maksimov, E. P. Lukashev, M. Moldenhauer, T. Friedrich and N. N. Sluchanko, Engineering the photoactive orange carotenoid protein with redox-controllable structural dynamics and photoprotective function, *Biochim. Biophys. Acta, Bioenerg.*, 2020, 148174.

25 N. N. Sluchanko, K. E. Klementiev, E. A. Shirshin, G. V. Tsoraev, T. Friedrich and E. G. Maksimov, The purple Trp288Ala mutant of Synechocystis OCP persistently quenches phycobilisome fluorescence and tightly interacts with FRP, *Biochim. Biophys. Acta, Bioenerg.*, 2017, **1858**, 1–11.

26 M. Gwizdala, A. Wilson, A. Omairi-Nasser and D. Kirilovsky, Characterization of the Synechocystis PCC 6803 Fluorescence Recovery Protein involved in photoprotection, *Biochim. Biophys. Acta, Bioenerg.*, 2013, **1827**, 348–354.

27 E. A. Shirshin, E. E. Nikonova, F. I. Kuzminov, N. N. Sluchanko, I. V. Elanskaya, M. Y. Gorbunov, V. V. Fadeev, T. Friedrich and E. G. Maksimov, Biophysical modeling of in vitro and in vivo processes underlying regulated photoprotective mechanism in cyanobacteria, *Photosynth. Res.*, 2017, **133**, 261–271.

28 H. Bao, M. R. Melnicki, E. G. Pawlowski, M. Sutter, M. Agostoni, S. Lechno-Yossef, F. Cai, B. L. Montgomery and C. A. Kerfeld, Additional families of orange carotenoid proteins in the photoprotective system of cyanobacteria, *Nat. Plants*, 2017, **3**, 17089.

29 M. Ludwig and D. A. Bryant, Transcription profiling of the model cyanobacterium Synechococcus sp. strain PCC 7002 by Next-Gen (SOLiD™) sequencing of cDNA, *Front. Microbiol.*, 2011, **2**, 41.

30 M. Sutter, A. Wilson, R. L. Leverenz, R. Lopez-Igual, A. Thurotte, A. E. Salmeen, D. Kirilovsky and C. A. Kerfeld,

Crystal structure of the FRP and identification of the active site for modulation of OCP-mediated photoprotection in cyanobacteria, *Proc. Natl. Acad. Sci. U. S. A.*, 2013, **110**, 10022–10027.

31 Y. B. Slonimskiy, E. G. Maksimov, E. P. Lukashev, M. Moldenhauer, C. M. Jeffries, D. I. Svergun, T. Friedrich and N. N. Sluchanko, Functional interaction of low-homology FRPs from different cyanobacteria with Synechocystis OCP, *Biochim. Biophys. Acta, Bioenerg.*, 2018, **1859**, 382–393.

32 N. N. Sluchanko, Y. B. Slonimskiy, E. A. Shirshin, T. Friedrich, E. G. Maksimov and M. Moldenhauer, OCP–FRP protein complex topologies suggest a mechanism for controlling high light tolerance in cyanobacteria, *Nat. Commun.*, 2018, **9**, 3869.

33 Y. Lu, H. Liu, R. G. Saer, H. Zhang, C. M. Meyer, V. L. Li, L. Shi, J. D. King, M. L. Gross and R. E. Blankenship, Native Mass Spectrometry Analysis of Oligomerization States of Fluorescence Recovery Protein and Orange Carotenoid Protein: Two Proteins Involved in the Cyanobacterial Photoprotection Cycle, *Biochemistry*, 2017, **56**, 160–166.

34 Y. Lu, H. Liu, R. Saer, V. L. Li, H. Zhang, L. Shi, C. Goodson, M. L. Gross and R. E. Blankenship, A Molecular Mechanism for Nonphotochemical Quenching in Cyanobacteria, *Biochemistry*, 2017, **56**, 2812–2823.

35 A. Thurotte, C. Bourcier de Carbon, A. Wilson, L. Talbot, S. Cot, R. López-Igual and D. Kirilovsky, The cyanobacterial Fluorescence Recovery Protein has two distinct activities: Orange Carotenoid Protein amino acids involved in FRP interaction, *Biochim. Biophys. Acta, Bioenerg.*, 2017, **1858**, 308–317.

36 M. Moldenhauer, N. N. Sluchanko, N. N. Tavraz, C. Junghans, D. Buhrke, M. Willoweit, L. Chiappisi, F.-J. Schmitt, V. Vukojević, E. A. Shirshin, V. Y. Ponomarev, V. Z. Paschenko, M. Gradzielski, E. G. Maksimov and T. Friedrich, Interaction of the signaling state analog and the apoprotein form of the orange carotenoid protein with the fluorescence recovery protein, *Photosynth. Res.*, 2018, **135**, 125–139.

37 N. N. Sluchanko, Y. B. Slonimskiy, M. Moldenhauer, T. Friedrich and E. G. Maksimov, Deletion of the short N-terminal extension in OCP reveals the main site for FRP binding, *FEBS Lett.*, 2017, **591**, 1667–1676.

38 E. G. Maksimov, M. Moldenhauer, E. A. Shirshin, E. A. Parshina, N. N. Sluchanko, K. E. Klementiev, G. V. Tsoraev, N. N. Tavraz, M. Willoweit, F. J. Schmitt, J. Breitenbach, G. Sandmann, V. Z. Paschenko, T. Friedrich and A. B. Rubin, A comparative study of three signaling forms of the orange carotenoid protein, *Photosynth. Res.*, 2016, **130**, 389–401.

39 S. Gupta, M. Sutter, S. G. Remesh, M. A. Dominguez-Martin, H. Bao, X. A. Feng, L.-J. G. Chan, C. J. Petzold, C. A. Kerfeld and C. Y. Ralston, X-ray radiolytic labeling reveals the molecular basis of orange carotenoid protein photoprotection and its interactions with fluorescence recovery protein, *J. Biol. Chem.*, 2019, **294**, 8848–8860.

40 F. Muzzopappa, A. Wilson and D. Kirilovsky, Interdomain interactions reveal the molecular evolution of the orange carotenoid protein, *Nat. Plants*, 2019, **5**, 1076–1086.

41 F. Muzzopappa, A. Wilson, V. Yogarajah, S. Cot, F. Perreau, C. Montigny, C. Bourcier de Carbon and D. Kirilovsky, The paralogs to the C-terminal domain of the cyanobacterial OCP are carotenoid donors to HCPs, *Plant Physiol.*, 2017, **175**, 1283–1303.

42 S. R. Bench, P. Heller, I. Frank, M. Arciniega, I. N. Shilova and J. P. Zehr, Whole genome comparison of six Crocosphaera watsonii strains with differing phenotypes, *J. Phycol.*, 2013, **49**, 786–801.

43 J. Yang, R. Yan, A. Roy, D. Xu, J. Poisson and Y. Zhang, The I-TASSER Suite: protein structure and function prediction, *Nat. Methods*, 2015, **12**, 7–8.

44 G. E. Crooks, G. Hon, J. Chandonia and S. E. Brenner, WebLogo: A Sequence Logo Generator, *Genome Res.*, 2004, **14**, 1188–1190.

45 H. Ashkenazy, S. Abadi, E. Martz, O. Chay, I. Mayrose, T. Pupko and N. Ben-Tal, ConSurf 2016: an improved methodology to estimate and visualize evolutionary conservation in macromolecules, *Nucleic Acids Res.*, 2016, **44**, W344–W350.

46 S. Kumar, G. Stecher, M. Li, C. Knyaz and K. Tamura, MEGA X: Molecular Evolutionary Genetics Analysis across Computing Platforms, *Mol. Biol. Evol.*, 2018, **35**, 1547–1549.

47 F. Sievers, A. Wilm, D. Dineen, T. J. Gibson, K. Karplus, W. Li, R. Lopez, J. D. Thompson, D. G. Higgins, H. Mcwilliam, M. Remmert and J. So, Fast, scalable generation of high-quality protein multiple sequence alignments using Clustal Omega, *Mol. Syst. Biol.*, 2011, **7**, 539.

